Deborup Sanyal

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Education

Syracuse University 2026 - Present
PhD, Computer Science & Engineering *Syracuse, NY*

University at Buffalo 2021 - 2023
MS, Computer Science & Engineering *Buffalo, NY*

Netaji Subhas Institute of Technology, University of Delhi 2016 - 2020
B.Eng, Computer Engineering *Delhi, India*

Research Interests

I'm broadly interested in computer systems research, especially in storage systems, memory systems, operating systems, and computer architecture. I'm currently exploring memory and storage systems that bridge hardware capabilities with the software stacks that use them. Specifically focused on redesigning host–device interfaces so that addresses carry performance meaning.

Areas: Disaggregated Memory, Processing-Near-Memory, Flash Memory and SSDs, File Systems, Storage Interfaces, Data-Intensive Systems

Technical Skills

Languages: Python, Java, C, C++, SQL, Scala, Bash/Shell, R, JavaScript, TypeScript, Groovy, Octave
Databases: SQL Server, Cockroach DB, MySQL, Oracle DB, PostgreSQL, Snowflake, Hive, DBeaver, Mongo DB
Machine Learning & Data Science: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, SciPy, MLFlow, OpenCV
Big Data & Cloud: Spark, Airflow, Kafka, Kubernetes, Hadoop, AWS, GCP, Docker, S3, MapR, Parquet, Arrow
Libraries, Frameworks & Tools: FastAPI, Spring Boot, REST, React.js, Next.js, Maven, SQLAlchemy, Alembic, Pydantic, Asyncio, Jenkins, Ansible, OpenTelemetry, Prometheus, Grafana, Terraform, Git, Linux, HTML, HashiCorp Vault, Autosys, SonarQube, Nexus Repository, Tableau, Solace, JupyterLab, Jira, Neo4j

Research/Industry Experience

Syracuse University Aug 2026 – Present
Research Assistant *Syracuse, NY*

- Designing an implicit quality-of-service interface where memory and storage addresses carry performance expectations. It spans SSD firmware, file system placement, and CXL fabric-manager and kernel allocator design.
- Building a profiling methodology that identifies the file system blocks applications actually stall on, using wait attribution and causal delay injection with device-mapper.
- [Mentor: Dr. Bryan S. Kim]

University of Texas at Dallas Jan 2026 – July 2026
Research Assistant *Richardson, TX*

- Leveraging LLMs to infer goal-state specifications in PDDL for QoS planning with numerical constraints.
- Exploring Relational Deep Learning training and inference directly in DBMS to optimize SQL queries.
- [Mentors: Dr. I-Ling Yen, Dr. Farokh Bastani]

Georgia Institute of Technology Mar 2025 – Jan 2026
Researcher, Center for Scientific Software Engineering *Atlanta, GA*

- Improved the performance and features of **grid2op ML framework**. Applied profiling to overcome memory bottlenecks, **enabling 2.4x speedup in serial runs** and heretofore inaccessible levels of parallel runs.
- Connected a higher-fidelity power computation backend in **Julia to the Python-based** ML software stack.

Nomura Holding America Inc. Apr 2023 – Jan 2025
Software Engineer - Data Platform, Chief Data Office *New York, NY*

- **Built** microservices for processing **10 million+** trades in real-time for **13+ data quality checks** of cash trades on Nomura's firmwide **trade front to back** initiative using **Java, Spark, Spring Boot, Kafka, Hive Metastore, Solace**.
- **Engineered real-time trade normalization system** transforming multi-source financial data **(FIX, XML)** into **standardized formats via Spark/Kafka pipeline**, deployed to deliver tailored data for different units reporting needs.

- **Saved cloud-hosting costs by utilizing AWS Lambda** to develop APIs, enabling efficient data processing with an ingestion layer using **Kafka Streaming** to handle trade messages and **dockerized** the downstream apps.
- Implemented complex **Spark RDD transformations** to process **real-time XML trade data streams**, enabling format validation and ensuring data integrity throughout the trade pipeline.
- Developed **high-performance rule-runner application using Scala** to process and validate 10 million+ daily trades across Cash, FX, and IRS portfolios.
- **Architected cloud-native MLOps platform** accelerating ML solutions delivery by **75%**, implementing robust connection, ingestion, feature engineering, model selection pipelines and governance frameworks.
- Engineered efficient **data storage solutions** by implementing distributed file systems using **Parquet and Arrow formats in HDFS on RHEL**, significantly optimizing data accessibility and processing speed.
- **Owned** backend for in-house **AutoML** application using **SQLAlchemy, MLFlow and Python libraries (Paramiko, Scikit-Learn, TensorFlow, BeautifulSoup)** to automate implementation of ML models across diverse **ML/AI** projects.
- Designed **data profiling system with anomaly detection, automated rule recommendation engine and interactive visualization components using Pandas, NumPy, and FastAPI**, reducing data quality issue identification by **25%**.
- Enhanced **Kubernetes** container orchestration for cross-cloud portability and resource utilization efficiency by **30%**.
- Increased **efficiency by 80%** by shifting critical apps on **Airflow for seamless orchestration.** Introduced observability using **OpenTelemetry Prometheus and Loki stack on Grafana Logs and Metrics**.
- **Led** the development and delivery of **Incidents Dashboard** to collect data, visualize it and send alerts for **7+ reports** using **ServiceNow API, Pivot Tables, React, SFTP** critical to save downtime on production services by **70%**.
- Standardized **CI/CD, infrastructure-as-code** of data applications by using **Git configuring Jenkins** to trigger **Ansible deploy on to target UNIX (RHEL) servers** automating deployment by **50%**.
- [Mentors: Mr. Anand Rao, Mr. Rakesh Ranjan and Mr. Santanu Sengupta]

University at Buffalo Jun 2022 – Aug 2022

Research Assistant, Data Intensive Distributed Computing Laboratory *Buffalo, NY*

- Optimized **metadata management by 15%** for end-to-end big data transfers by building **APIs to expose metadata** for the **OneDataShare** project in collaboration with NSF and IBM.
- Worked on integrating generation and saving of **SSL-keys for CockroachDB** of various users using **HashiCorp vault**.
- [Mentor: Dr. Tevfik Kosar]

ALT Digital Technologies Pvt. Ltd. Aug 2020 – Jun 2021

Software Engineer *Noida, India*

- Built web app of e-commerce client using **Java, Oracle CX** and reduced **compilation time by 60%**.
- **Optimized SQL queries** to cut API response rate by **75%**. Personalized shopping cart using **deep learning** models.
- [Mentor: Mr. Vikas Goyal]

Illinois Institute of Technology May 2019 - Aug 2020

Research Intern, Bhushan Research Group *Chicago, IL*

- Developed comprehensive analysis pipeline for **Alzheimer's disease prediction** using large-scale biomarker data.
- Implemented **missing data imputation methods** including stochastic regression and kNN using NumPy and scikit-learn libraries.
- Created specialized preprocessing procedures for **RNA sample quality filtering, dataset normalization and differential gene (DEG) analysis** using significant analysis of microarrays (SAM) method.
- Applied multiple ML methods (**Logistic Regression, Lasso Regression, SVMs, Random Forest, Neural Networks**) for predictive modeling.
- Developed **custom feature engineering** techniques to handle high-dimensionality data challenges.
- [Mentor: Dr. Abhinav Bhushan]

Petroleum Technology Co. (Petrotec) Dec 2017 – Jan 2018

IT Intern *Doha, Qatar*

- Helped in efforts maintaining Windows Server setups, Active Directory, and Microsoft Entra ID.
- [Mentor: Mr. Ranjith Paul]

Qatar Computing Research Institute May 2017 – Aug 2017

Research Intern, Social Computing *Doha, Qatar*

- Developed a full pipeline of data processing: the data collection from **Facebook Ads API at a large scale**, correcting sampling biases using **Iterative Proportional Fitting model**, and finally visualization using the **Tableau system**.
- This development consisted of **iterative examination of the data**, its comparison to outside sources such as **census and other demographic statistics**, and assessing the quality and stability of the dataset.
- [Mentors: Dr. Yelena Mejova and Dr. Ingmar Weber]

Research Projects

Transport Protocol Chat Application | *C++, Network Simulation* Dec 2022

- Implemented three transport protocols **(Alternating-Bit, Go-Back-N, Selective-Repeat)** with comprehensive performance analysis framework.
- Evaluated protocol throughput under varying network conditions including **packet loss probabilities (0.1-0.8) and window sizes (10-500 packets).**
- Demonstrated **Go-Back-N's superior performance** through quantitative benchmarking and analyzed trade-offs between throughput, latency, and reliability.

COVID-19 Prediction Using CT-Scan Analysis | *Python, TensorFlow, Medical Imaging* May 2022

- Developed deep learning models for COVID-19 severity prediction using lung CT-scan analysis with **DenseNet-121 and DenseNet-169 architectures.**
- Achieved **75.58% accuracy** in predicting COVID-19 severity from STOIC2021 dataset.
- Conducted comparative analysis demonstrating **DenseNet-121's superior performance over DenseNet-169** for medical image classification using PyTorch framework.

Robust Classifiers for Datasets with Imbalance and Noise | *Python, JupyterLab, Pattern Recognition* Dec 2021

- Developed learning systems based on **SVM and Logistic Regression** as ML methods and **LDAM-DRW and Symmetric Loss (SL)** as Deep Learning methods.
- Proposed novel **gamma distribution-based resampling technique** for minority class to improve generalization without sacrificing performance on majority classes.
- Enhanced model performance on imbalanced MNIST dataset with noisy labels through comprehensive comparative analysis.

Deep Reinforcement Learning for Stock Trading | *Python, JupyterLab, Reinforcement Learning* Dec 2021

- Implemented Deep RL-based stock trading system leveraging historical market data to optimize trading decisions.
- Designed environment supporting 3 discrete actions (buy, sell, hold) while managing investment capital and portfolio value.
- Utilized **Q-learning** with factors including shares held, stock value, and total portfolio value.

Multi-agent Hide and Seek Game | *Python, JupyterLab, Reinforcement Learning* Nov 2021

- Developed multi-agent reinforcement learning environment for emergent competitive behavior.
- Implemented both **Actor-Critic and Deep Q-Network algorithms** to train agents in different scenarios.
- Compared performance of different RL approaches through comprehensive analysis.

Map Image to Image Translation Using Conditional GANs | *Python, Computer Vision* Dec 2019

- Implemented **Conditional Generative Adversarial Networks** for image translation tasks.
- Explored computer vision techniques for image-to-image translation applications.

Posters/Talks/Presentations

COMPLECS: Intermediate Linux and Shell Scripting, San Diego Supercomputing Center	Jul 2025
COMPLECS: Code Migration, San Diego Supercomputing Center	Jun 2025
COMPLECS: Using Regular Expressions with Linux Tools, San Diego Supercomputing Center	May 2025
LEEP Demo Day 2025, Lawrence Berkeley National Laboratory	May 2025
COMPLECS: Data Storage and File Systems, San Diego Supercomputing Center	May 2025
PSC Machine Learning and BIG DATA Workshop, Georgia Institute of Technology	May 2025
COMPLECS: Data Transfer, San Diego Supercomputing Center	Mar 2025
Computer Science & Engineering Graduate Demo Days, University at Buffalo	Dec 2021
Artificial Intelligence Projects Workshop, Netaji Subhas Institute of Technology	Jan 2020
UG Summer Research Immersion Expo, Illinois Institute of Technology	Jul 2019

Honors & Awards

Most Valuable Team Award in Q4 FY'24, Nomura	Mar 2024
Best Reinforcement Learning project in the CSE Demo Days, University at Buffalo	Dec 2021
Final Project selected in the top 5 teams for Pattern Recognition	Dec 2021
Member of organizing committee for Towards Extensible and Adaptable Methods in Computing Conference	Mar 2018
Hosted annual inter-college freshers quiz event, Abhimanyu, NSIT	Sep 2017
Qualified the Joint Entrance Examination, Govt. of India	Jun 2016
Awarded Gold Medal for exceptional academic performance for 7 consecutive school years	May 2016
AP Scholar for 5/5 in Calculus BC and Physics C: E&M	Apr 2016
800/800 on SAT Subject Tests in Mathematics Level 2, Physics and Chemistry	Oct 2015
Selected for the GCIMUN-2014 Student Conference at the United Nations Headquarters	May 2014
Gold Medals in level 2 of SOF's International Mathematics and National Science Olympiads	Jan 2014
Obtained the DELF A1 and A2 French language certifications by Education Dept., France	Apr 2013